Shivakshit Patri

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Summary

I approach technical challenges with a first principles mindset, and I like to build scalable, automated systems. I am fortunate to have made an impact at places like Amazon, MIT Media Labs, Synopsys, & Intel. But in June 2024, I stepped away from the workforce to address my chronic illnesses. This break was dedicated to navigating significant personal health and family matters, which required my full attention. Now that they are completely resolved, I return to the field with a renewed sense of purpose and a sharpened focus, eager to apply my technical discipline to mission-critical initiatives.

Experience

- **Amazon** — Bellevue, Washington, United States
  Computer Vision Software Engineer, Data Center Security — *Oct, 2021 - June, 2024*
  - **Deep Learning**: Trained deep learning models (*YoloV5, U-Net)* to more than 90% precision & recall performance for human detection and image segmentation on surveillance camera footage from 135 Amazon data centers.
  - **Geometric Computer Vision**: Employed geometric algorithms and techniques (*Hough-transform, SIFT & Stereo-triangulation*) for edge detection and depth estimation.
  - **Data Engineering**: Built *Amazon RDS* tables from model inference dump in *S3* buckets using ETL jobs in *AWS Glue*. Designed dashboards in *AWS Quicksight* to monitor performance metrics calculated in *AWS Athena*.
  - **Data Collection, & Pre-Processing**: Automated repetitive workflows like collection, segmentation & labeling of petabytes of video data using *AWS Kinesis, EC2, Sagemaker, SQS, Lambda & S3.*
  - **Infrastructure Management**: Migrated from manual AWS Console resource configuration to Infrastructure as Code using *AWS Cloud Development Kit (CDK)* & *CloudFormation*, improving maintainability, enabling programmatic management, and streamlining deployment processes.
  - **DevOps & MLOps**: Placed alarms & interactive dashboards to track key production metrics (p95 latency, 4xx/5xx errors). Optimized deployment workflow by refining dependencies—reducing mean deployment time from 10 to 6 minutes.
  - **Mentor**: Mentored three junior developers/interns to onboard with the team's workflow to quickly familiarize them with our tech stack & software architecture.
  - **Planning & Facilitation**: Performed SCRUM master duties for daily stand-ups, sprint-planning, retrospectives and reviews.
- **Arizona State University** — Tempe, Arizona, United States
  Graduate Research Assistant, Gautam Dasarathy's Research Group — *August, 2019 - December, 2020*
  - **Active Learning on Graphs**: Surveyed top state-of-the-art active learning algorithms for graph label prediction problems. By leveraging ASU's High Performance Computing Cluster, I implemented, analyzed and compared the performance of these prediction algorithms on synthetic data using *Networkx* python library & *Jupyter notebooks*.
- **MIT Media Labs & L.V. Prasad Eye Institute** — Hyderabad, Telangana, India
  Computer Vision Intern, Center for Innovation — *April, 2017 - October, 2017*
  - **Ophthalmoscope**: Developed image stitching software with *OpenCV* and *PyGame* on *Raspberry Pi* for an eye imaging device.
  - **Glaucoma Screening Device**: Led a mult-disciplinary team of 10 to build a portable *Raspberry Pi* powered 3D-printed, glaucoma detection device using intraocular pressure analysis, reducing risk of vision loss by 90%.
- **Synopsys Inc.** — Hyderabad, Telangana, India
  Application Engineer, Process Development Kit Team — *June, 2016 - January, 2017*
  - **Custom Compiler**: Wrote *Python* logic for the electronic design automation tool used by major chip makers.
- **Intel** — Bengaluru, Karnataka, India
  Circuit Design Intern, Analog & Mixed Signal Design Team — *July, 2015 - December, 2015*
  - **Analog to Digital Converter**: Designed a sigma-delta ADC for temperature sensing in 14nm technology node.
- **Supreme Industries** — Jalgaon, Maharasthra, India
  Computer Vision Intern — *May, 2014 - July, 2014*
  - **Pipe counting software in MATLAB**: Reduced time for counting pipes by 100x using Hough-Transform.

Education

- **Arizona State University** — Tempe, Arizona, United States
  Master of Science in Computer Engineering — *Aug, 2018 – Aug, 2021*
- **Birla Institute of Technology and Science, Pilani** — Hyderabad, Telangana, India
  Bachelor of Engineering in Electronics & Communications — *Aug, 2012 – July, 2016*

Skills

- **Programming & Markup Languages**: *C/C++, Python, Java, Javascript, Typescript, SQL, MATLAB, Bash, HTML, CSS, LaTeX.*
- **Cloud Microservices**: *AWS EC2, S3, Lambda, RDS, DynamoDB, IAM, SNS, SQS, Glue, Athena, QuickSight, CloudFormation, CloudWatch, API Gateway, Sagemaker, CloudBuild, Kinesis, Step Functions.*
- **Libraries**: (Python) *Collections, Numpy, LangChain, LangGraph, JAX, Pytorch, Tensorflow, Keras, Scikit-learn, Toon, Scipy, Matplotlib, Seaborn, FastAPI, Pandas, Plotly, Pillow, Boto3, NLTK, OpenCV, Networkx, Django, Flask, Pytest, Pygame, PyQt, Beautiful Soup*; (Java) *Log4j, JUnit*; (C++) *Standard Template Library, OpenCV.*
- **Platforms & Frameworks**: *Git, Docker, Kubernetes, Hugging Face, Kagglehub, Spark, Hadoop, Perforce, OpenClaw.*
- **Software Tools**: *Visual Studio, PyCharm, Jupyter, Colab, Microsoft Office, Teams, Slack, Copilot, Codex, Gemini, Claude code.*
- **OS**: *MacOS, Windows, Ubuntu, RHEL, Raspberry Pi OS.*

Academic Publications

- S. Paramkusham, **P. Shivakshit**, K. M. M. Rao and B.V.V.S.N.P. Rao. (2015). "A new features extraction method based on polynomial regression for the assessment of breast lesion Contours", *2015 International Conference on Industrial Instrumentation and Control (ICIC), Pune, India.*

Other Achievements

- **4th Hack Harvard Hackathon (2018)**: Developed an online tool in Harvard University's annual hackathon, that scrapes image, video and text data to investigate the presence of child abuse.
- **5th Engineering the Eye Hackathon (2017)**: Won the **"Best team"** award in the hackathon hosted by MIT for developing a medical imaging device for early detection of glaucoma.

Languages

English: Proficient   **Hindi**: Proficient   **Telugu**: Native